

Adversarial Fairness

Qualitative & Analytic Report

Dataset: HIMS-real10k

2026-06-27

Accuracy: 87.8% -> 81.2%
Iterations: 19
Gender: PASS — P-rule 87% $\geq$ 80%
region_origin: PASS — P-rule 81% $\geq$ 80%
educ_level: PASS — P-rule 81% $\geq$ 80%

Qualitative analysis (LLM)

The disadvantaged group for 'Gender' is Male, who have a predicted-positive rate 16.6% lower than Females (80.0% vs 69.4%), resulting in a P-rule of 86.8%. The model amplifies the actual outcome-rate gap between genders, with Females having an actual outcome rate 34.2% higher than Males (98.0% vs 63.8%). For 'region_origin', the disadvantaged group is Center-West, who have a predicted-positive rate 15.3% lower than Greater Tunis (81.8% vs 66.5%), resulting in a P-rule of 81.3%. The model mirrors the actual outcome-rate gap between regions, with Greater Tunis having an actual outcome rate 41.1% higher than Center-West (91.9% vs 50.8%). Debiaseding improved the P-rule from 62.5% to 81.3%, but at a cost of reducing accuracy from 87.8% to 81.2%. The gaps in predicted-positive rates are not designed, but rather flow through correlated proxies such as education level and region of origin.

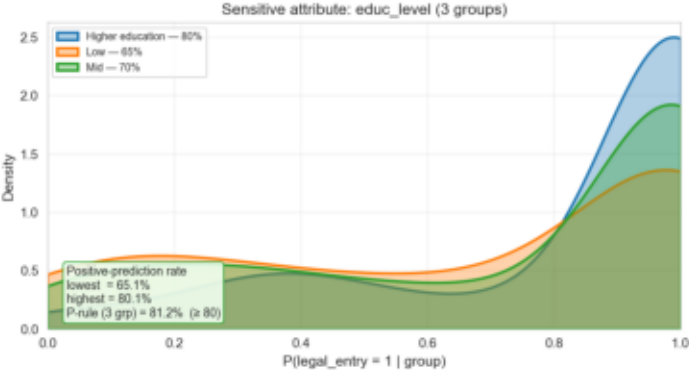
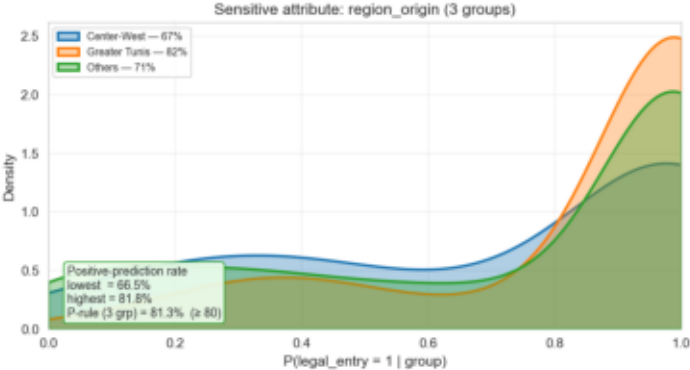
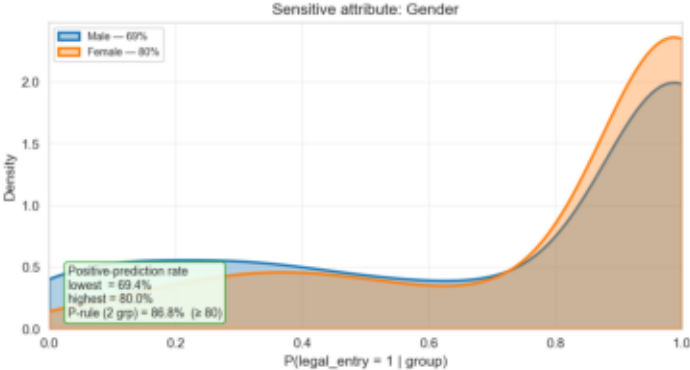
Fairness metrics (before -> after)

attribute	P-rule	DP	EoDD	EoOpp
Gender	67.0 -> 86.8	31.6 -> 10.6	8.6 -> 11.1	8.7 -> 8.4
region_origin	62.5 -> 81.3	17.3 -> 6.2	0.5 -> 6.8	0.7 -> 7.1
educ_level	62.2 -> 81.2	26.5 -> 11.1	12.0 -> 4.2	8.3 -> 2.2

P-rule higher is fairer; DP / EoDD / EoOpp lower is fairer.

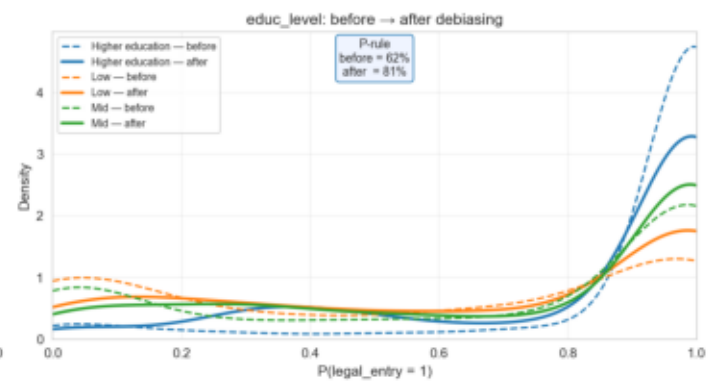
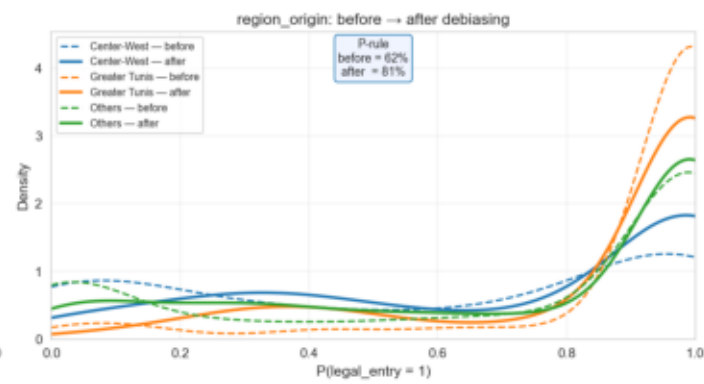
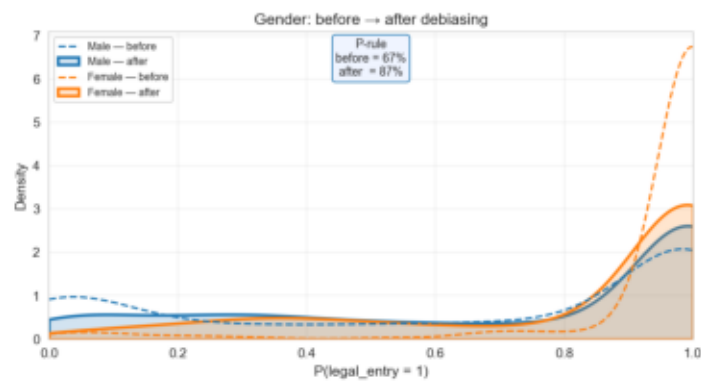
Predicted-outcome distribution by group

HIMS-real10k — predicted $P(\text{legal_entry} = 1)$ by sensitive group



Before → after debiasing

HIMS-real10k — before → after debiasing



Equalized odds — before vs after
HIMS-real10k — equalized odds before vs after

